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(54) **LAVATORY FAUCET WITH CENTER WATER INLET**

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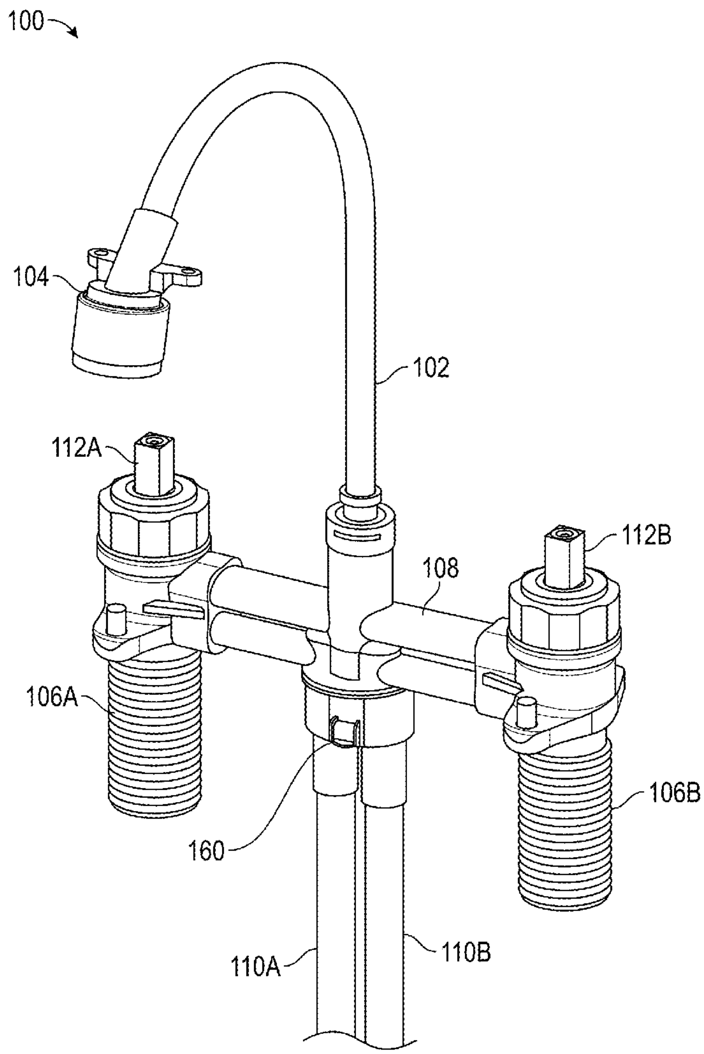
(57) **ABSTRACT**

A faucet assembly includes a bridge having a central water inlet configured to removably connect to a cold water supply line and a hot water supply line; a cold water valve connected to the bridge at a first end of the bridge and configured to receive cold water from a lower cold water pathway; and a hot water valve connected to the bridge at a second end of the bridge and configured to receive hot water from a lower hot water pathway, wherein the bridge includes a mixing chamber for receiving cold water from the cold water valve via an upper cold water pathway and hot water from the hot water valve via an upper hot water pathway. The faucet assembly may have a central extended shank or extended side shanks extending below the cold water valve and the hot water valve respectively.



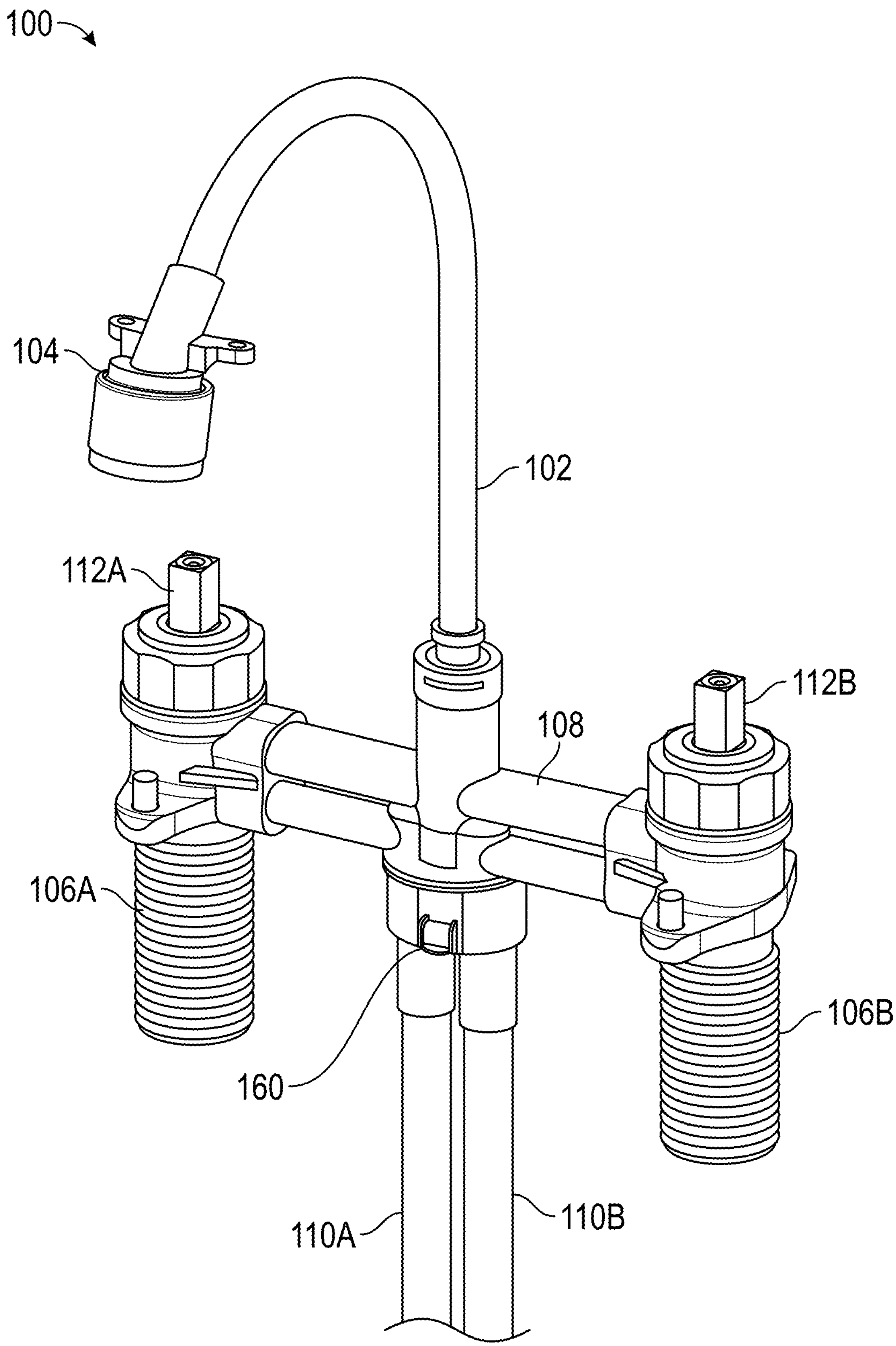


FIG. 1

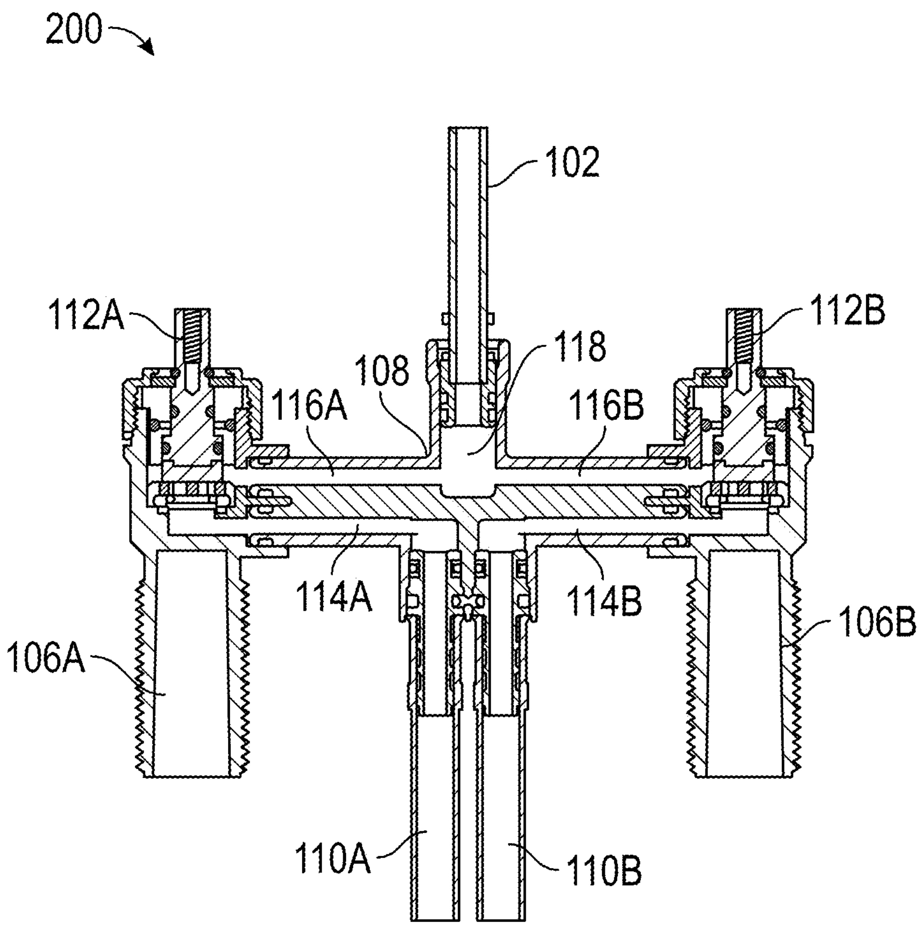

200
102
112A
112B
118
108
116A
116B
114A
114B
106A
106B
110A
110B

FIG. 2

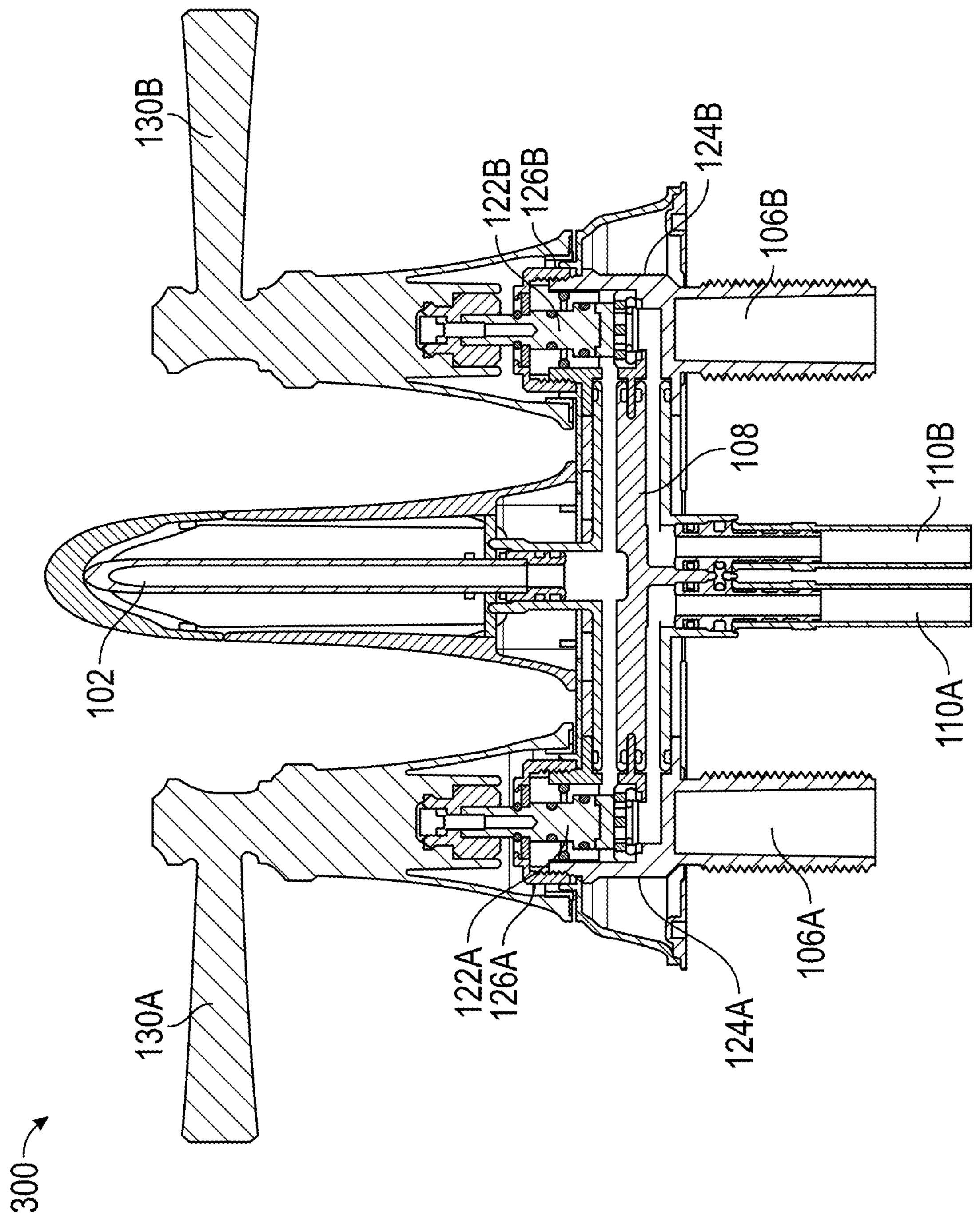

300
130A
130B
122A
126A
122B
126B
124A
124B
106A
106B
102
108
110A
110B

FIG. 3

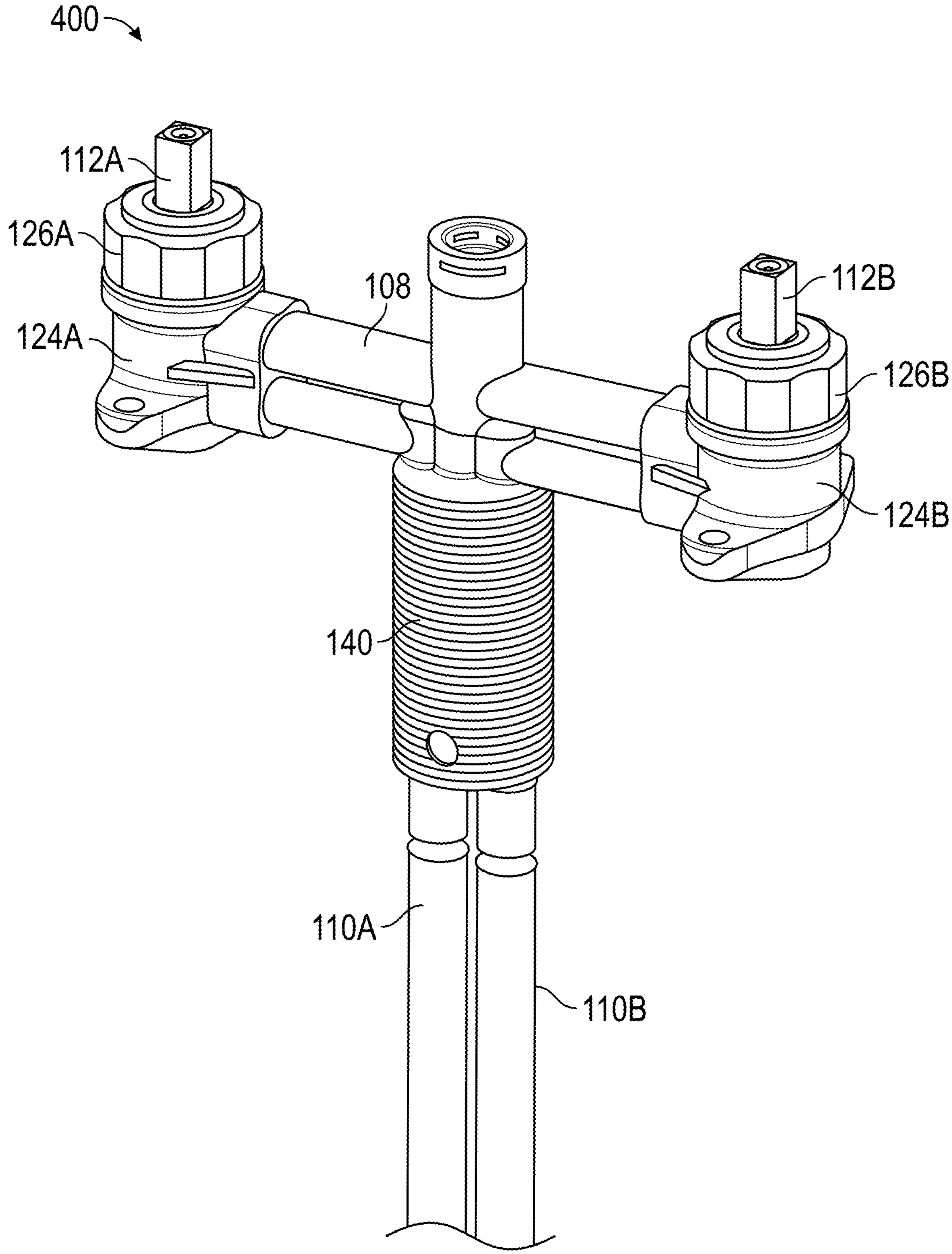


FIG. 4

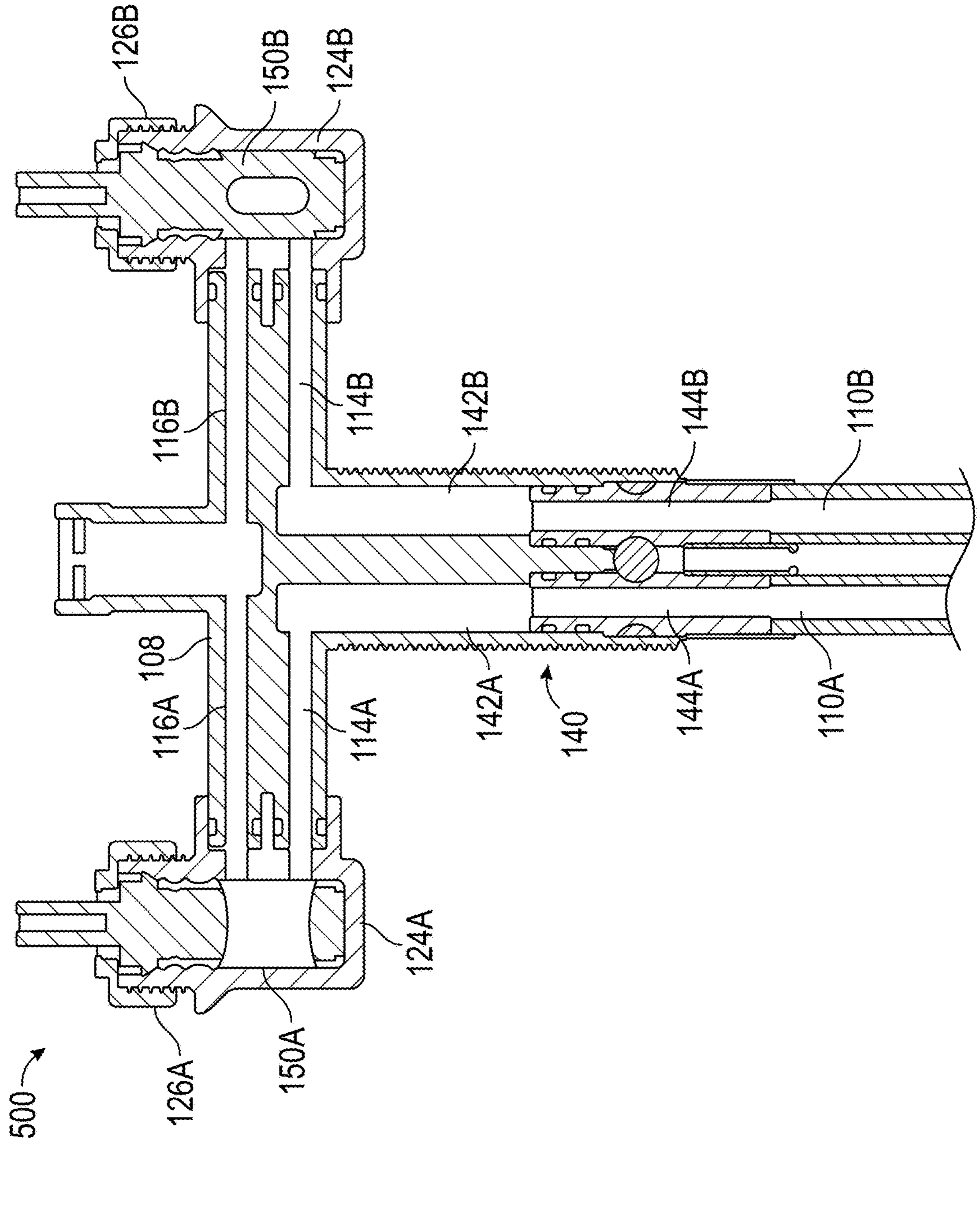

500
126A
150A
124A
126B
150B
124B
116B
114B
142B
144B
110B
108
116A
114A
142A
140
144A
110A

FIG. 5

LAVATORY FAUCET WITH CENTER WATER INLET

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of U.S. Provisional Application No. 63/558,362, filed Feb. 27, 2024, the entire contents of which is incorporated herein by reference.

FIELD

[0002] The present disclosure generally relates to lavatory faucets, and more particularly, to lavatory faucets having a center water inlet.

BACKGROUND

[0003] A dual control lavatory faucet includes a hot water control and a cold water control. Generally, within each control resides a cartridge or valve that controls the flow rate of the water flowing to the cartridge from a water supply line. The relative position of the cartridge controls the flow rate of the water flowing from the water supply line, through the cartridge, and to a mixing chamber, which receives water from both a hot water supply line (via a hot water cartridge) and a cold water supply line (via a cold water cartridge). From the mixing chamber, the mixed temperature water flows through an outlet tube and out of the faucet outlet.

[0004] Dual control lavatory faucet assemblies are configured to connect to the water supply lines (i.e., hot water supply line and cold water supply line) directly at their respective cartridges. Therefore, the cold water supply line connects the main water supply to the faucet assembly at the cold water cartridge of the assembly, and the hot water supply line connects the main water supply to the faucet assembly at the hot water cartridge of the assembly. Dual control faucets require three openings in the sink deck or other mounting surface to connect the faucet assembly to the hot and cold water supply lines.

SUMMARY

[0005] Provided herein are faucet assemblies comprising a center water inlet. Unlike conventional dual control lavatory faucet assemblies, which receive a cold water inlet at the cold water cartridge and a hot water inlet at the hot water cartridge (i.e., through the side openings of the sink deck or mounting surface), the faucet assemblies described herein receive both the hot water inlet and the cold water inlet at a central location, or through the central opening of the sink deck or mounting surface to which the faucet assembly is configured to mount. This configuration allows the dual faucet assembly to be mounted to a single-opening sink deck or mounting surface (i.e., it doesn't require three openings). Additionally, in some embodiments, the faucet assembly includes the cold and hot water supply lines, preventing the user from having to purchase the water supply lines separately as is required with conventional faucet assemblies.

[0006] To accommodate the central water inlet, the faucet assemblies described herein include a bridge that connects the water supply lines, cartridges, and water outlet. Specifically, the water supply lines are configured to couple to the bridge. Hot water flows from a hot water supply line to the bridge, and from the bridge to the hot water cartridge. From the hot water cartridge, the hot water flows to a mixing chamber, mixes with the cold water, and then flows out through the water outlet. Similarly, cold water flows from the cold water inlet to the bridge, and from the bridge to the cold water cartridge. From the cold water cartridge, the cold water flows to a mixing chamber, mixes with the hot water, and then flows out through the water outlet.

[0007] A faucet assembly as described herein comprises one or more extended shanks. These extended shanks are configured to pass through an opening in the sink deck or mounting surface to which the faucet assembly can be mounted. The extended shank(s) can be threaded, such that a nut can be coupled to the extended shank, securing the faucet assembly to the sink deck or mounting surface. In some embodiments, a faucet assembly as described herein may comprise a central extended shank. In some embodiments, a faucet assembly may comprise extended side shanks. In some embodiments, a faucet assembly may comprise both extended side and central shanks. A faucet assembly with only one extended shank (i.e., a central extended shank) allows a dual control faucet assembly to be mounted to a sink deck or mounting surface with only one opening, whereas conventional dual control faucet assemblies generally require three openings in the sink deck or mounting surface to be mounted.

[0008] Another benefit of the faucet assemblies provided herein is that they are provided with the water supply lines. In many conventional faucet assemblies, a user is required to purchase the water supply lines separately from the faucet assembly. However, as provided herein, the water supply lines can be included as part of the faucet assembly, reducing the total number of parts that the user needs to source and purchase.

[0009] In some embodiments, provided is a faucet assembly, the faucet assembly comprising: a bridge comprising a central water inlet configured to removably connect to a cold water supply line and a hot water supply line; a cold water valve housed within a cold water valve housing comprising an extended shank configured to pass through a first opening in a mounting surface, wherein the cold water valve housing is connected to the bridge at a first end of the bridge and the cold water valve is configured to receive cold water from a lower cold water pathway in the bridge; and a hot water valve housed within a hot water valve housing comprising an extended shank configured to pass through a second opening in the mounting surface, wherein the hot water valve housing is connected to the bridge at a second end of the bridge and the hot water valve is configured to receive hot water from a lower hot water pathway in the bridge, wherein the bridge comprises a mixing chamber configured to receive cold water from the cold water valve via an upper cold water pathway in the bridge and hot water from the hot water valve via an upper hot water pathway in the bridge.

[0010] In some embodiments of the faucet assembly, the bridge does not comprise a central extended shank configured to pass through a central opening in the mounting surface.

[0011] In some embodiments of the faucet assembly, the lower cold water pathway, the lower hot water pathway, the upper cold water pathway, and the upper hot water pathway are each horizontal.

[0012] In some embodiments of the faucet assembly, the upper cold water pathway is configured to transport cold water in a first direction and the lower cold water pathway is configured to transport cold water in a second direction, wherein the first direction and the second direction are opposite.

[0013] In some embodiments of the faucet assembly, the upper hot water pathway is configured to transport water in a third direction and the lower hot water pathway is configured to transport water in a fourth direction, wherein the third direction and the fourth direction are opposite.

[0014] In some embodiments of the faucet assembly, the upper cold water pathway is directly above the lower cold water pathway, and the upper hot water pathway is directly above the lower hot water pathway.

[0015] In some embodiments of the faucet assembly, the bridge comprises an engineering thermoplastic.

[0016] In some embodiments of the faucet assembly, the engineering thermoplastic is selected from the group consisting of polyamides, polyesters, polycarbonates, acrylonitrile-butadiene-styrene, polysulfones (PSU), polyethersulfones (PESU), cyclic olefin copolymer (COC), acrylonitrile-styrene-acrylate (ASA), polyphenylene oxides (PPO), polyphenylene sulfides (PPS), polyphenylenesulfones (PPSU), polyether ether ketones (PEEK), polyethylenimine (PEI), polyphthalamides (PPA), polyacetals, copolymers thereof, and blends thereof.

[0017] In some embodiments of the faucet assembly, the bridge comprises a glass-filled polyamide or a glass-filled polyphthalamide.

[0018] In some embodiments of the faucet assembly, the bridge comprises an injection molded thermoplastic.

[0019] In some embodiments of the faucet assembly, the bridge comprises at least one of an engineering thermoplastic, a glass-filled polyamide, a glass-filled polyphthalamide, or an injection molded thermoplastic.

[0020] In some embodiments of the faucet assembly, the mounting surface comprises the first opening for the extended shank of the cold water valve housing, the second opening for the extended shank of the hot water valve housing, and a central opening for the cold water supply line and the hot water supply line.

[0021] In some embodiments of the faucet assembly, the cold water supply line and the hot water supply line are configured to connect to the bridge at a location that is below the mixing chamber of the bridge.

[0022] In some embodiments of the faucet assembly, the faucet assembly comprises an outlet tube configured to removably connect to the bridge and receive mixed water from the mixing chamber.

[0023] In some embodiments, provided is a faucet assembly comprising: a bridge comprising a central extended shank comprising a central water inlet configured to removably connect to a cold water supply line and a hot water supply line; a cold water valve connected to the bridge at a first side and configured to receive cold water from a lower cold water pathway in the bridge; and a hot water valve connected to the bridge at a second side and configured to receive hot water from a lower hot water pathway in the bridge, wherein the bridge comprises a mixing chamber configured to receive cold water from the cold water valve via an upper cold water pathway in the bridge and hot water from the hot water valve via an upper hot water pathway in the bridge.

[0024] In some embodiments of the faucet assembly, the central extended shank of the bridge comprises a cold water pathway connecting the cold water supply line to the lower cold water pathway of the bridge and a hot water pathway connecting the hot water supply line to the lower hot water pathway of the bridge.

[0025] In some embodiments of the faucet assembly, the cold water valve is housed within a cold water valve housing that does not comprise an extended shank, and the hot water valve is housed within a hot water valve housing that does not comprise an extended shank.

[0026] In some embodiments of the faucet assembly, the lower cold water pathway, the lower hot water pathway, the upper cold water pathway, and the upper hot water pathway are each horizontal.

[0027] In some embodiments of the faucet assembly, the upper cold water pathway is configured to transport cold water in a first direction and the lower cold water pathway is configured to transport cold water in a second direction, wherein the first direction and the second direction are opposite.

[0028] In some embodiments of the faucet assembly, the upper hot water pathway is configured to transport water in a third direction and the lower hot water pathway is configured to transport water in a fourth direction, wherein the third direction and the fourth direction are opposite.

[0029] In some embodiments of the faucet assembly, the upper cold water pathway is directly above the lower cold water pathway, and the upper hot water pathway is directly above the lower hot water pathway.

[0030] In some embodiments of the faucet assembly, the bridge comprises an engineering thermoplastic.

[0031] In some embodiments of the faucet assembly, the engineering thermoplastic is selected from the group consisting of polyamides, polyesters, polycarbonates, acrylonitrile-butadiene-styrene, polysulfones (PSU), polyethersulfones (PESU), cyclic olefin copolymer (COC), acrylonitrile-styrene-acrylate (ASA), polyphenylene oxides (PPO), polyphenylene sulfides (PPS), polyphenylenesulfones (PPSU), polyether ether ketones (PEEK), polyethylenimine (PEI), polyphthalamides (PPA), polyacetals, copolymers thereof, and blends thereof.

[0032] In some embodiments of the faucet assembly, the bridge comprises a glass-filled polyamide or a glass-filled polyphthalamide.

[0033] In some embodiments of the faucet assembly, the bridge comprises an injection molded thermoplastic.

[0034] In some embodiments of the faucet assembly, the faucet assembly is configured to mount to a mounting surface comprising a single opening through which the central extended shank of the bridge is configured to pass through and a mounting surface comprising three openings, wherein the central extended shank of the bridge is configured to pass through a central opening of the three openings.

[0035] In some embodiments of the faucet assembly, the cold water supply line and the hot water supply line are configured to connect to the bridge at a location that is below the mixing chamber of the bridge.

[0036] In some embodiments of the faucet assembly, the faucet assembly comprises an outlet tube configured to removably connect to the bridge and receive mixed water from the mixing chamber.

[0037] In some embodiments, any one or more of the features, characteristics, or elements discussed above with respect to any of the embodiments may be incorporated into any of the other embodiments mentioned above or described elsewhere herein.

BRIEF DESCRIPTION OF THE FIGURES

[0038] FIG. **1** shows a perspective view of a faucet assembly having a central water inlet, according to some embodiments;

[0039] FIG. **2** shows a cross-sectional view of a faucet assembly having a central water inlet and comprising extended side shanks, according to some embodiments;

[0040] FIG. **3** shows a cross-sectional view of a dual control faucet comprising a faucet assembly having a central water inlet and extended side shanks, according to some embodiments;

[0041] FIG. **4** shows a perspective view of a faucet assembly having a central water inlet and comprising a central extended shank, according to some embodiments; and

[0042] FIG. **5** shows a cross-sectional view of a faucet assembly having a central water inlet and comprising a central extended shank, according to some embodiments.

DETAILED DESCRIPTION

[0043] Provided are faucet assemblies comprising a central water inlet. Conventional dual control faucet assemblies receive the hot water supply line and the cold water supply line at opposing ends of the faucet assembly (i.e., below the hot control and cold control, respectively). However, the faucet assemblies provided receive both the hot water supply line and the cold water supply line centrally, or below the outlet tube of a dual control faucet. This configuration allows a dual control faucet assembly to be mounted on a sink deck or mounting surface that only has one opening (i.e., is configured for a single control faucet).

[0044] In some embodiments, the faucet assemblies provided herein are provided with the water supply lines coupled to the faucet assembly. This can enable a user to more easily purchase and install the faucet assembly. Specifically, conventional dual control faucet assemblies require additionally sourcing and purchasing the water supply lines. However, in this case, the water supply lines can be included with the faucet assembly for purchase. Additionally, having the water supply lines already coupled to the faucet assembly allows a user to more easily install the faucet assembly. The water supply lines (already coupled to the faucet assembly) can be easily threaded down through a single (or central) opening in the sink deck or mounting surface. The opposing ends of the water supply lines (those not coupled to the faucet assembly) simply need to be coupled to the main water supply. Without having the water supply lines already coupled to the faucet assembly, a user generally has to reach up underneath the sink deck to couple the water supply lines to the faucet assembly. This can be uncomfortable and difficult to do.

[0045] FIG. **1** shows a perspective view of a lavatory faucet with center water inlet **100**, according to some embodiments. The lavatory faucet with center water inlet **100** includes an outlet tube **102**, faucet outlet **104**, cold water shank **106**A, hot water shank **106**B, bridge **108**, cold water supply line **110**A, hot water supply line **110**B, cold water valve stem **112**A, hot water valve stem **112**B, and metal clip **160**.

[0046] As shown, lavatory faucet with center water inlet **100** is a dual control faucet, meaning that the faucet is controlled with two separate controls—a cold water control and a hot water control. The cold water control comprises cold water valve stem **112**A, cold water shank **106**A, and cold water supply line **110**A. The hot water control comprises hot water valve stem **112**B, hot water shank **106**B, and hot water supply line **110**B. In a conventional dual control faucet, the water supply lines (i.e., cold water supply line **110**A and hot water supply line **110**B) are coupled to the faucet assembly at the water valves. Thus, in a conventional dual control faucet, the cold water supply line **110**A would be connected directly to the cold water valve (e.g., through the cold water shank **106**A) and the hot water supply line **110**B would be connected directly to the hot water valve (e.g., through the hot water shank **106**B).

[0047] However, as shown in FIG. **1**, this is not the case in the disclosed lavatory faucet with center water inlet **100**. Instead, the cold water supply line **110**A and the hot water supply line **110**B are connected to the cold water valve and the hot water valve indirectly through the bridge **108** of the lavatory faucet with center water inlet **100**. The bridge **108** comprises waterways directing cold water from the cold water supply line **110**A to the cold water valve, and from the cold water valve to a mixing chamber within the bridge **108** where the cold water mixes with hot water, and from the mixing chamber to the outlet tube **102** and faucet outlet **104**. The bridge **108** also comprises waterways directing hot water from the hot water supply line **110**B to the hot water valve, and from the hot water valve to the mixing chamber where the hot water mixes with the cold water, and from the mixing chamber to the outlet tube **102** and faucet outlet **104**. By having both the cold water supply line **110**A and the hot water supply line **110**B connect to the rest of the faucet assembly at a central location, this allows for the faucet assembly to be mounted to a surface deck having only a single opening. Traditionally, surface decks having only one opening can only accommodate a single-control or a single handle faucet assembly. However, the configuration described herein (i.e., dual-control faucet assembly with center water inlet) allows a dual-control faucet assembly to be mounted to a surface deck with only a single opening.

[0048] The two water supply lines (i.e., cold water supply line **110**A and hot water supply line **110**B) are connected to the bridge **108** via a fastening mechanism. In some embodiments, the fastening mechanism may be a metal clip, as shown in FIG. **1** as item **160**. This metal clip secures the cold water supply line **110**A to the bridge **108** and the hot water supply line **110**B to the bridge **108**. A fastening mechanism such as metal clip **160** allows the water supply lines to be securely fastened, yet easily unfastened at the same time (for replacement, installation, etc.).

[0049] In some embodiments, the outlet tube **102** may be configured to removably attach to the bridge **108**. The outlet tube **102** can receive mixed water from a mixing chamber of the bridge **108**.

[0050] FIG. **2** shows a cross-sectional view of a faucet assembly **200** having a central water inlet and comprising extended side shanks, according to some embodiments. As shown, faucet assembly **200** includes outlet tube **102**, cold water shank **106**A, hot water shank **106**B, bridge **108**, cold water supply line **110**A, hot water supply line **110**B, cold water valve stem **112**A, and hot water valve stem **112**B. Bridge **108** comprises lower cold water pathway **114**A, lower hot water pathway **114**B, upper cold water pathway **116**A, upper hot water pathway **116**B, and mixing chamber **118**. In some embodiments, the faucet assembly **200** may comprise extended side shanks, but not a central extended shank.

[0051] The specific faucet assembly configuration shown in FIG. **2** can be mounted to a surface deck having three openings. The cold water supply line **110**A and hot water supply line **110**B are configured to pass through a central opening in a surface deck. The cold water shank **106**A and hot water shank **106**B are both extended shanks, meaning that they are each configured to pass though a side opening in a surface deck having three openings. The extended shanks are designed to secure the faucet assembly **200** to the surface deck. In some embodiments, the cold water shank **106**A and hot water shank **106**B are threaded to allow for mounting and easily securing to the mounting surface.

[0052] Faucet assembly **200** is configured to receive cold water at cold water supply line **110**A. Cold water supply line **110**A is configured to connect to bridge **108** at one end (as shown) and a water supply at an opposing end. From the water supply, the cold water passes through the cold water supply line **110**A vertically upwards and enters the bridge **108** at a central location (with respect to the faucet assembly **200**) on a lower side of the bridge **108**. Once in the bridge **108**, the cold water runs horizontally through the lower cold water pathway **114**A and to the cold water valve inlet (in a direction away from a center of the faucet assembly). The cold water valve stem **112**A may be controlled by a user control (e.g., handle, knob). The specific rotation location of the cold water valve stem **112**A controls the amount of cold water (or flow rate of cold water) that enters and exits the cold water valve. From the cold water valve, the cold water exits at the cold water valve outlet and reenters the bridge **108** at the upper cold water pathway **116**A. The upper cold water pathway **116**A directs the cold water back in to the center of the faucet assembly **200**. Specifically, cold water flows from the cold water valve outlet horizontally through the upper cold water pathway **116**A and to a mixing chamber **118**. The mixing chamber is designed to receive both the cold water and hot water and mix them to form mixed temperature water. The mixing chamber **118** can be located at a central location (with respect to the faucet assembly **200**). In some embodiments, the mixing chamber may be located directly above the central water inlet (i.e., the location at which the cold water supply line **110**A and the hot water supply line **110**B connect to the bridge **108**). In some embodiments, the mixing chamber **118** is located directly below the outlet tube **102**.

[0053] Faucet assembly **200** is configured to receive hot water at hot water supply line **110**B. Hot water supply line **110**B is configured to connect to bridge **108** at one end (as shown) and a water supply at an opposing end. From the water supply, the hot water passes through the hot water supply line **110**B vertically upwards and enters the bridge **108** at a central location (with respect to the faucet assembly **200**) on a lower side of the bridge **108**. Once in the bridge **108**, the hot water runs horizontally through the lower hot water pathway **114**B and to the hot water valve inlet (in a direction away from a center of the faucet assembly). The hot water valve stem **112**B may be controlled by a user control (e.g., handle, knob). The specific rotation location of the hot water valve stem **112**B controls the amount of hot water (or flow rate of hot water) that enters and exits the hot water valve. From the hot water valve, the hot water exits at the hot water valve outlet and reenters the bridge **108** at the upper hot water pathway **116**B. The upper hot water pathway **116**B directs the hot water back in to the center of faucet assembly **200**. Specifically, hot water flows from the hot water valve outlet horizontally through the upper hot water pathway **116**B and to a mixing chamber **118**. The mixing chamber is designed to receive both the hot water and cold water and mix them to form mixed temperature water.

[0054] Each upper water path in bridge **108** may be directly above its corresponding lower water path. Therefore, in some embodiments, this means that the upper cold water pathway **116**A is directly above the lower cold water pathway **114**A, and similarly, upper hot water pathway **116**B is directly above lower hot water pathway **114**B. In some embodiments, the cold water running through upper cold water pathway **116**A runs in a direction that is directly opposite the direction that the cold water in lower cold water pathway **114**A runs. Similarly, in some embodiments, the hot water running through upper hot water pathway **116**B runs in a direction that is directly opposite the direction that the hot water in lower hot water pathway **114**B runs.

[0055] As described above, the mixing chamber **118** receives cold water from upper cold water pathway **116**A and hot water from upper hot water pathway **116**B. The hot water and the cold water in mixing chamber **181** mixes to form mixed temperature water. From the mixing chamber **118**, the mixed temperature water is directed to a conduit in the outlet tube **102** and out through the faucet outlet.

[0056] The cold water supply line **110**A and the hot water supply line **110**B can be configured to connect to the bridge **108** at a location that is directly below a mixing chamber **118** of the bridge **108**.

[0057] FIG. **3** shows another cross-sectional view of a dual control faucet comprising a faucet assembly **300** having a central water inlet and extended side shanks, according to some embodiments. Specifically, FIG. **3** shows the faucet assembly with user controls **130**A (cold water) and **130**B (hot water). Also shown in FIG. **3** are the cold water supply line **110**A, hot water supply line **110**B, cold water valve housing **124**A, hot water valve housing **124**B, cold water valve **122**A, hot water valve **122**B, locking nut **126**A for cold water valve **122**A, locking nut **126**B for hot water valve **122**B, outlet tube **102**, cold water user control **130**A, and hot water user control **130**B. As shown, like the configuration shown in FIG. **2**, the configuration here also includes cold water shank **106**A and hot water shank **106**B that are extended, meaning that they can extend through side openings in a mounting surface having three openings (i.e., designed for a conventional dual-control faucet assembly).

[0058] As shown and described in more detail with respect to FIG. **2**, cold water enters the faucet assembly at cold water supply line **110**A and passes to the bridge **108**. From the bridge **108**, the cold water enters cold water valve **122**A which is situated within cold water valve housing **124**A. Cold water valve housing **124**A can be connected to bridge **108** at a first end. Cold water valve **122**A is held in place with cold water locking nut **126**A. The position of the stem of the cold water valve **122**A is controlled by a user manipulating cold water user control **130**A and dictates how much cold water enters and exits cold water valve **122**A (i.e., a flow rate of the cold water flowing through the cold water valve **122**A). From the cold water valve **122**A, the cold water is directed back into the bridge **108** where it mixes with hot water in a mixing chamber, and then passes to the outlet tube **102** and out of the faucet outlet.

[0059] Similarly, as shown and described in more detail with respect to FIG. **2**, hot water enters the faucet assembly at hot water supply line **110**B and passes to the bridge **108**.

From the bridge **108**, the hot water enters hot water valve **122**B which is situated within hot water valve housing **124**B. Hot water valve housing **124**B can be connected to bridge **108** at a second end. Hot water valve **122**B is held in place with hot water locking nut **126**B. The position of the stem of the hot water valve **122**B is controlled by a user manipulating hot water user control **130**B and dictates how much hot water enters and exits hot water valve **122**B (i.e., a flow rate of the hot water flowing through the hot water valve **122**B). From the hot water valve **122**B, the hot water is directed back into the bridge **108** where it mixes with cold water in a mixing chamber, and then passes to the outlet tube **102** and out of the faucet outlet.

[0060] FIG. **4** shows a perspective view of a faucet assembly **400** having a central water inlet and comprising a central extended shank, according to some embodiments. A difference between this configuration shown here in FIG. **4** and the configurations shown elsewhere (e.g., FIG. **2**, FIG. **3**) is that the faucet assembly **400** includes a central shank **140** that is extended, or configured to pass through a central opening of a mounting surface. This configuration does not include side shanks that secure the faucet assembly to side openings. Therefore, the faucet assembly configuration shown in FIG. **4** can be mounted to a mounting surface having a single opening or a mounting surface having three openings (and configured for conventional dual-control faucet assemblies which require three openings). This allows more flexibility, since the dual-control faucet assembly **400** having only a central shank **140** that is extended (and not including extended side shanks) can be mounted to a mounting surface having either a single opening or three openings. In some embodiments, faucet assembly **400** may comprise a central extended shank but not extended side shanks.

[0061] Faucet assembly **400** of FIG. **4** includes cold water supply line **110**A and hot water supply line **110**B that are configured to connect to central shank **140**. Central shank **140** is shown as a central extended shank, as described above, and can include two connections, one that receives a cold water supply line **110**A and one that receives a hot water supply line **110**B.A waterway through the central shank **140** allows water to pass from the cold water supply line **110**A and to the bridge **108**, and similarly, a second waterway through the central shank **140** allows water to pass from the hot water supply line **110**B and to the bridge **108**.

[0062] Faucet assembly **400** also includes cold water valve housing **124**A and hot water valve housing **124**B. Cold water valve housing **124**A houses a cold water valve of the faucet assembly **400**, and hot water valve housing **124**B houses a hot water valve of the faucet assembly **400**. The cold water valve is held in place with a locking nut **126**A and the hot water valve is held in place with locking nut **126**B. Finally, FIG. **4** shows the cold water valve stem **112**A and the hot water valve stem **112**B extending upwards from the cold water valve housing **124**A and locking nut **126**A, and hot water valve housing **124**B and locking nut **126**B, respectively.

[0063] FIG. **5** shows a cross-sectional view of a faucet assembly **500** having a central water inlet and comprising a central extended shank, according to some embodiments.

[0064] Faucet assembly **500** is configured to receive cold water from a cold water supply at cold water supply line **110**A. From cold water supply line **110**A, the cold water passes to cold water pathway **142**A of central shank **140**. As shown, cold water supply line **110**A is removably connected to the central shank **140** via connection **144**A. From the cold water pathway **142**A, the cold water flows to the bridge **108**. In the bridge, the cold water flows to lower cold water pathway **114**A, through which the cold water flows horizontally to the cold water valve **150**A. Cold water valve **150**A is controlled by a user at a cold water control (e.g., knob, handle). The amount of the cold water that flows into the cold water valve **150**A directly depends on the relative location (e.g., rotation) of the cold water valve control. From the cold water valve **150**A, the cold water flows to an upper cold water pathway **116**A in the bridge **108**. From the upper cold water pathway **116**A in the bridge, the cold water flows in to a mixing chamber, where the cold water mixes with hot water from a hot water flow path.

[0065] Faucet assembly **500** is configured to receive hot water from a hot water supply at hot water supply line **110**B. From hot water supply line **110**B, the hot water passes to hot water pathway **142**B of the central shank **140**. As shown, hot water supply line **110**B is removably connected to the central shank **140** via connection **144**B. From the hot water pathway **142**B, the hot water flows to the bridge **108**. In the bridge, the hot water flows to lower hot water pathway **114**B, through which the hot water flows horizontally to the hot water valve **150**B. Hot water valve **150**B is controlled by a user at a hot water control (e.g., knob, handle). The amount of hot water that flows into the hot water valve **150**B directly depends on the relative location (e.g., rotation) of the hot water valve control. From the hot water valve **150**B, the hot water flows to an upper hot water pathway **116**B in the bridge **108**. From the upper hot water pathway **116**B in the bridge, the hot water flows into a mixing chamber, where the hot water mixes with cold water from the cold water pathway.

[0066] In some embodiments, lower cold water pathway **114**A and lower hot water pathway **114**B are configured to transport water horizontally away from a central location in the bridge **108**. In some embodiments, upper cold water pathway **116**A and upper hot water pathway **116**B are configured to transport water horizontally toward a central location of the bridge **108**. In some embodiments, cold water pathway **142**A and hot water pathway **142**B are configured to transport water vertically upwards to the bridge. In some embodiments, the upper cold water pathway **116**A and the lower cold water pathway **114**A are configured to transport cold water in opposite directions. In some embodiments, the upper hot water pathway **116**B and the lower hot water pathway **114**B are configured to transport hot water in opposite directions.

[0067] Bridge **108** may comprise engineering thermoplastics. In some embodiments, the engineering thermoplastics may include but is not limited to polyamides, polyesters, polycarbonates, acrylonitrile-butadiene-styrene, polysulfones (PSU), polyethersulfones (PESU), cyclic olefin copolymer (COC), acrylonitrile-styrene-acrylate (ASA), polyphenylene oxides (PPO), polyphenylene sulfides (PPS), polyphenylenesulfones (PPSU), polyether ether ketones (PEEK), polyethylenimine (PEI), polyphthalamides (PPA), polyacetals, copolymers thereof, or blends thereof. In some embodiments, bridge **108** may comprise a glass-filled polyamide or a glass-filled polyphthalamide. In some embodiments, bridge **108** may comprise an injection molded thermoplastic.

[0068] As shown in FIG. **5**, cold water valve housing **124**A houses cold water valve **150**A, and locking nut **126**A holds cold water valve **150**A in place. Similarly, hot water valve housing **124**B houses hot water valve **150**B, and locking nut **126**B holds hot water valve **150**B in place.

[0069] As shown herein, the cold water pathways of the faucet assembly are shown on the left side of the assembly, and the hot water pathways of the faucet assembly are shown on the right side of the assembly (from the perspective of a user using the faucet assembly). However, this configuration can also be presented in an alternative configuration, where the hot water pathways are on the left, and the cold water pathways are on the right.

[0070] The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

[0071] Although the disclosure and examples have been fully described with reference to the accompanying figures, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims. Finally, the entire disclosure of the patents and publications referred to in this application are hereby incorporated herein by reference.

[0072] Any of the systems, methods, techniques, and/or features disclosed herein may be combined, in whole or in part, with any other systems, methods, techniques, and/or features disclosed herein.

1. A faucet assembly comprising:

a bridge comprising a central water inlet configured to removably connect to a cold water supply line and a hot water supply line;

a cold water valve housed within a cold water valve housing comprising an extended shank configured to pass through a first opening in a mounting surface, wherein the cold water valve housing is connected to the bridge at a first end of the bridge and the cold water valve is configured to receive cold water from a lower cold water pathway in the bridge; and

a hot water valve housed within a hot water valve housing comprising an extended shank configured to pass through a second opening in the mounting surface, wherein the hot water valve housing is connected to the bridge at a second end of the bridge and the hot water valve is configured to receive hot water from a lower hot water pathway in the bridge,

wherein the bridge comprises a mixing chamber configured to receive cold water from the cold water valve via an upper cold water pathway in the bridge and hot water from the hot water valve via an upper hot water pathway in the bridge.

2. The faucet assembly of claim **1**, wherein the upper cold water pathway is configured to transport cold water in a first direction and the lower cold water pathway is configured to transport cold water in a second direction, wherein the first direction and the second direction are opposite.

3. The faucet assembly of claim **1**, wherein the upper hot water pathway is configured to transport water in a third direction and the lower hot water pathway is configured to transport water in a fourth direction, wherein the third direction and the fourth direction are opposite.

4. The faucet assembly of claim **1**, wherein the upper cold water pathway is directly above the lower cold water pathway, and the upper hot water pathway is directly above the lower hot water pathway.

5. The faucet assembly of claim **1**, wherein the bridge comprises an engineering thermoplastic selected from the group consisting of polyamides, polyesters, polycarbonates, acrylonitrile-butadiene-styrene, polysulfones (PSU), polyethersulfones (PESU), cyclic olefin copolymer (COC), acrylonitrile-styrene-acrylate (ASA), polyphenylene oxides (PPO), polyphenylene sulfides (PPS), polyphenylenesulfones (PPSU), polyether ether ketones (PEEK), polyethylenimine (PEI), polyphthalamides (PPA), polyacetals, copolymers thereof, and blends thereof.

6. The faucet assembly of claim **5**, wherein the bridge comprises a glass-filled polyamide or a glass-filled polyphthalamide.

7. The faucet assembly of claim **1**, wherein the mounting surface comprises the first opening for the extended shank of the cold water valve housing, the second opening for the extended shank of the hot water valve housing, and a central opening for the cold water supply line and the hot water supply line.

8. The faucet assembly of claim **1**, wherein the cold water supply line and the hot water supply line are configured to connect to the bridge at a location that is below the mixing chamber of the bridge.

9. The faucet assembly of claim **1**, comprising an outlet tube configured to removably connect to the bridge and receive mixed water from the mixing chamber.

10. A faucet assembly comprising:

a bridge comprising a central extended shank comprising a central water inlet configured to removably connect to a cold water supply line and a hot water supply line;

a cold water valve connected to the bridge at a first side and configured to receive cold water from a lower cold water pathway in the bridge; and

a hot water valve connected to the bridge at a second side and configured to receive hot water from a lower hot water pathway in the bridge,

wherein the bridge comprises a mixing chamber configured to receive cold water from the cold water valve via an upper cold water pathway in the bridge and hot water from the hot water valve via an upper hot water pathway in the bridge.

11. The faucet assembly of claim **10**, wherein the central extended shank of the bridge comprises a cold water pathway connecting the cold water supply line to the lower cold water pathway of the bridge and a hot water pathway connecting the hot water supply line to the lower hot water pathway of the bridge.

12. The faucet assembly of claim **10**, wherein the cold water valve is housed within a cold water valve housing that does not comprise an extended shank, and the hot water valve is housed within a hot water valve housing that does not comprise an extended shank.

13. The faucet assembly of claim **10**, wherein the upper cold water pathway is configured to transport cold water in a first direction and the lower cold water pathway is configured to transport cold water in a second direction, wherein the first direction and the second direction are opposite.

14. The faucet assembly of claim **10**, wherein the upper hot water pathway is configured to transport water in a third direction and the lower hot water pathway is configured to transport water in a fourth direction, wherein the third direction and the fourth direction are opposite.

15. The faucet assembly of claim **10**, wherein the upper cold water pathway is directly above the lower cold water pathway, and the upper hot water pathway is directly above the lower hot water pathway.

16. The faucet assembly of claim **10**, wherein the bridge comprises an engineering thermoplastic selected from the group consisting of polyamides, polyesters, polycarbonates, acrylonitrile-butadiene-styrene, polysulfones (PSU), polyethersulfones (PESU), cyclic olefin copolymer (COC), acrylonitrile-styrene-acrylate (ASA), polyphenylene oxides (PPO), polyphenylene sulfides (PPS), polyphenylene-sulfones (PPSU), polyether ether ketones (PEEK), polyethylenimine (PEI), polyphthalamides (PPA), polyacetals, copolymers thereof, and blends thereof.

17. The faucet assembly of claim **16**, wherein the bridge comprises a glass-filled polyamide or a glass-filled polyphthalamide.

18. The faucet assembly of claim **10**, wherein the faucet assembly is configured to mount to a mounting surface comprising a single opening through which the central extended shank of the bridge is configured to pass through and a mounting surface comprising three openings, wherein the central extended shank of the bridge is configured to pass through a central opening of the three openings.

19. The faucet assembly of claim **10**, wherein the cold water supply line and the hot water supply line are configured to connect to the bridge at a location that is below the mixing chamber of the bridge.

20. The faucet assembly of claim **10**, comprising an outlet tube configured to removably connect to the bridge and receive mixed water from the mixing chamber.

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